

Course Code	MCA40150				
Course Category	Program Major				
Course Title	Advanced Database Management System				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3				3
Credits	3				3
Assessment Schema Code	TT1				
<u>Prerequisites:</u>					
Database Management System and Relational database management system, SQL					
<u>Course Objectives:</u>					
1. Understanding parallel database principles, distributed, XML, Objective oriented and mobile databases.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to: 1. Design parallel and distributed databases. 2. To work on mobile, XML databases.					
<u>Course Contents:</u>					
Unit 1: Database System Architectures (6 Sessions)					
Database System Architectures Centralized and Client-Server Architectures, Server System Architectures, Parallel Systems, Distributed Systems.					
Unit 2: Parallel and Distributed Databases (12 Sessions)					
Introduction, I/O Parallelism, Inter and Intra Query Parallelism, Inter and Intra operation Parallelism, Distributed Data Storage, Distributed Transactions, Commit Protocols, Concurrency Control, Distributed Query Processing – Three Tier Client Server Architecture.					
Unit 3: Objected Oriented databases (9 Sessions)					

Concepts for Object Databases, Object Identity, Object structure and Type Constructors, Encapsulation of Operations, Methods and Persistence, Type and Class Hierarchies and Inheritance, Complex Objects, Object Database Standards, Languages and Design: ODMG Model – ODL – OQL – Object Relational and Extended – Relational Systems: Object Relational features in SQL/Oracle

Unit 4: XML databases (9 Sessions)

XML Data Model, XML Documents, DTD and XML Schema, XML Querying, Web Databases – JDBC – Information Retrieval – Data Warehousing – Data Mining

Unit 5: Mobile databases & Multimedia databases (9 Sessions)

Location and Handoff Management - Effect of Mobility on Data Management - Location Dependent Data Distribution - Mobile Transaction Models - Concurrency Control - Transaction Commit Protocols - Mobile Database Recovery Schemes Multidimensional Data Structures – Image Databases – Text/Document Databases - Video Databases – Audio Databases – Multimedia Database Design.

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises based on the above units and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. Avi Silberschatz, Henry F. Korth, S. Sudarshan, “Database System Concept”, 4th Edition, 2001, ISBN: 0072283637, McGraw-Hill publications.
2. Ramez Elmasri, Shamkant Navathe, “Fundamental of Database Systems”, 5th Edition, 2009, ISBN: 9780321369574, Pearson Education.
3. Alexis Leon, Mathews Leon, “Database Systems concept”, 2002, ISBN: 9788125911654, Leon Vikas.

Supplementary

Reading: Web

Resources:

1. <https://www.tutorialspoint.com> ,
2. www.w3schools.com

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc.

https://onlinecourses.nptel.ac.in/noc22_cs47/preview

Pedagogy:

- Power point presentations
- Video



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| <ul style="list-style-type: none">● Demonstrations● Interactive discussion.● Practical based learning,● Project based learning, |
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